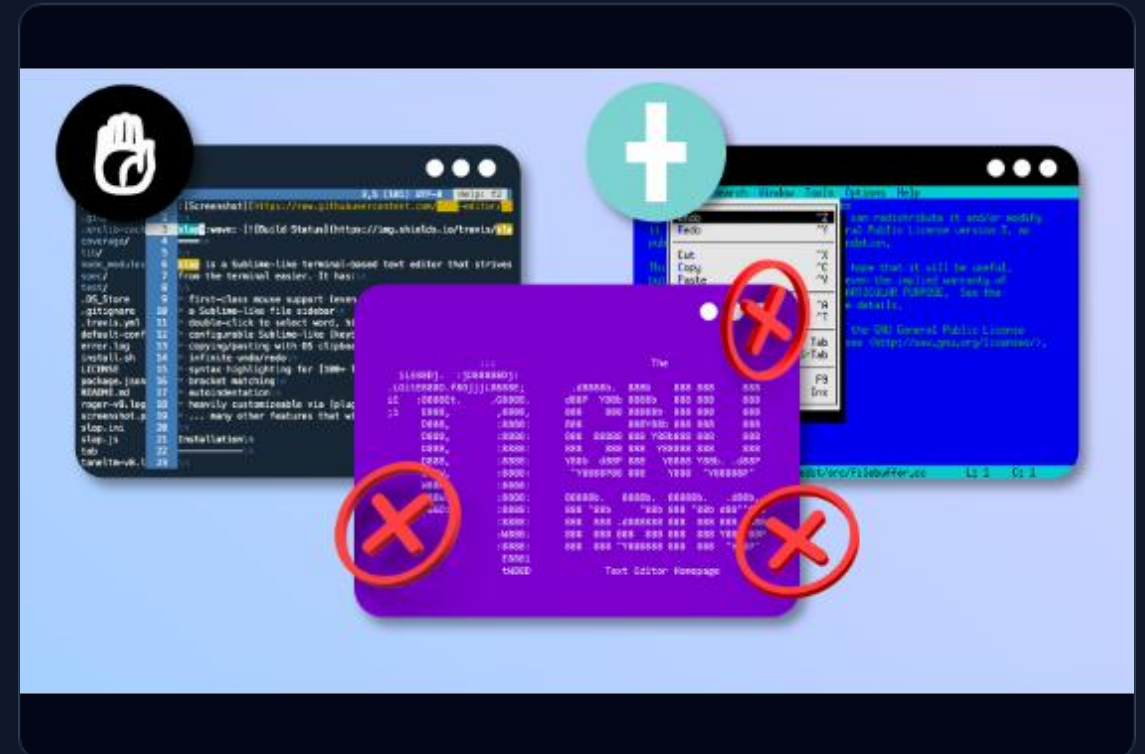


Module 04 (a)

Text Editing with nano

Why nano? Text Editing in Shell

- **Pre-installed Everywhere:** Available out of the box across almost all Linux distributions (Ubuntu, Debian, CentOS).
- **Beginner Friendly:** Unlike modal editors (Vim/Emacs), nano provides immediate modeless text insertion.
- **On-Screen Shortcuts:** Key combinations are constantly displayed at the bottom of the buffer for quick reference.
- **Essential for SSH Work:** Perfect for editing remote configuration files, C/Python scripts, and system logs.



Opening & Creating Files

Basic Usage

Open or create a file directly from terminal:

```
$ nano main.c
```

Jump to a specific line number directly on launch:

```
$ nano +25 script.py
```

Useful for jumping straight to compiler syntax errors!

Essential Flags

Show line numbers automatically:

```
$ nano -l config.txt
```

Open in read-only / view mode:

```
$ nano -v /etc/hosts
```

Create backup copies before saving changes:

```
$ nano -B file.txt
```

The Shell Session

Terminal Efficiency

nano operates completely inside your terminal session, eliminating any dependency on a graphical desktop environment (GUI).

Mastering keyboard-driven workflows allows seamless control over remote servers and embedded devices like Raspberry Pi.

bash — 109x34

```
TIME CMD
0:00.19 -/bin/zsh
0:00.19 -/bin/zsh
0:33.00 hugo serve
0:00.13 -/bin/zsh
0:01.11 /bin/zsh
0:04.73 /bin/zsh
0:01.12 /bin/zsh
0:01.08 /bin/zsh
0:00.22 /bin/zsh -i
0:00.20 -/bin/zsh
0:00.30 /bin/zsh -il
0:28.45 claude
0:00.65 npm exec chrome-devtools-mcp@latest
0:00.38 node /Users/nls/.npm/_npx/15c61037b1978c83/node_modules/.bin/chrome-
0:00.11 -zsh
0:00.01 bash
0:01.07 /bin/zsh
0:00.93 /bin/zsh
0:01.05 /bin/zsh
0:01.12 /bin/zsh
```

Key Notations & Basics



The ^ Notation

The caret symbol ^ denotes the **Ctrl** key. For instance, ^X means press Ctrl + X together.



The M- Notation

The letter M- represents the **Alt** (Meta) key (or Esc on macOS). E.g., M-U means Alt + U.



Direct Editing

There are no separate mode switches. Just start typing directly into the active terminal cursor position.

Core Commands at a Glance

A large, bold, red graphic of the nano command, consisting of a caret (^) followed by the letter O.

Write Out (Save)

Saving and Exiting Safely

Press Ctrl + O to save ("Write Out") modifications to disk. Press Enter to confirm the existing file name or type a new name to "Save As".

Press Ctrl + X to exit nano. If you have unsaved changes, nano will prompt you with Y (Yes) or N (No) before quitting.

Clipboard & Selection Operations

 **Ctrl + K** — Cut current line into the cutbuffer (press repeatedly to cut multiple consecutive lines).

 **Ctrl + U** — Uncut / Paste the contents of the cutbuffer at the current cursor location.

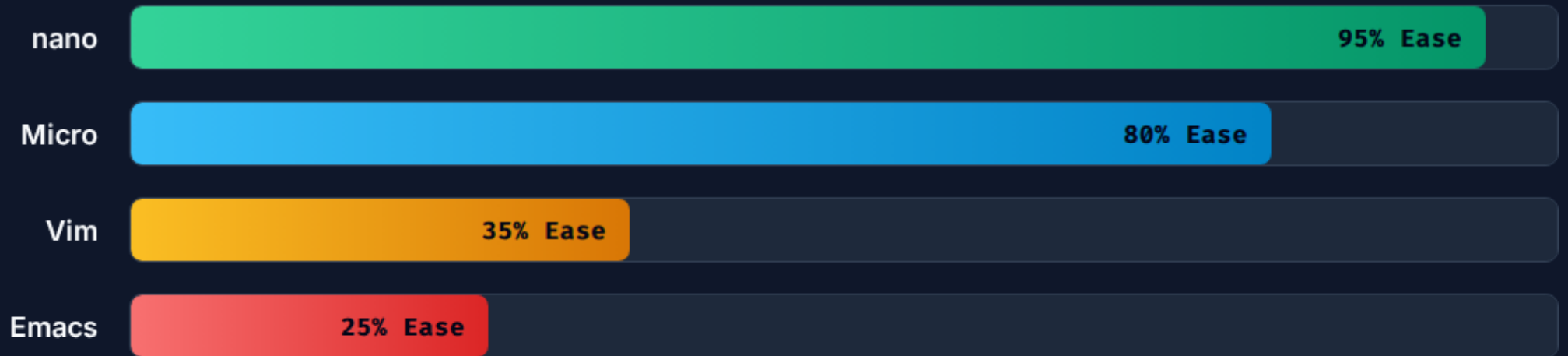
 **Alt + 6** — Copy current line (or selected block of text) without deleting it.

 **Alt + A** — Set / Unset selection mark; use arrow keys to highlight precise text regions.

Search & Navigation Reference

Shortcut	Function	Engineering Use Case
Ctrl + W	Where Is (Search)	Find specific variable or function definition
Alt + W	Search Next	Iterate through next matching log/code occurrences
Ctrl + \	Search & Replace	Refactor variable names across an entire script
Ctrl + _	Go to Line Number	Jump directly to line flagged by GCC compiler error
Ctrl + A / E	Start / End of line	Navigate quickly across long code commands

CLI Text Editors Benchmark



Comparison of Beginner Accessibility & Onboarding Speed across Linux Editors. Nano leads in zero-setup speed for server edits.

Customizing nano via .nanorc

The Config File

Location: ~/.nanorc

Configure persistent defaults so you don't need to pass command-line flags every time you launch nano.

```
$ nano ~/.nanorc
```

Recommended Configuration

Add these lines to your ~/.nanorc:

```
set linenumbers
```

```
set tabsize 4
```

```
set autoindent
```

```
set softwrap
```

Questions & Lab Session

Ready to practice? Launch your terminal and open your first file:

```
$ nano lab1_exercise.c
```

Image Sources



<https://static0.howtogeekimages.com/wordpress/wp-content/uploads/2025/12/a-comparison-of-terminal-text-editors-showing-modern-options-like-slap-and-tilde-as-superior-to-the-crossed-out-nano-editor.png?w=1600&h=900&fit=crop>

Source: www.howtogeek.com



<https://mintcdn.com/upsun-c9761871/eXV27YLeu5IESGxN/images/posts/insights/my-terminal-setup-mac-linux/uglybash.webp?fit=max&auto=format&n=eXV27YLeu5IESGxN&q=85&s=47d2e8a3fa5f6fd0a2f0e2190e3de0b6>

Source: developer.upsun.com